

PetalEdge Playbook

PetalEdge takes a team from raw camera or sensor data to a trained model running on real hardware — one open-source platform covering the whole pipeline, not a demo of one slice of it. Use this to pitch, demo, and answer the questions prospects actually ask.

INTERNAL — NOT CUSTOMER-FACING COLLATERAL

Why teams pick PetalEdge

Lead with these — each is a real, shipped capability, not a roadmap promise.

Open source, end to end

The whole pipeline — data, training, deployment — with nothing behind a black box.

SAY THIS

You're not locked into a closed platform you can't inspect or extend.

Three detection trainers, not one

In-house YOLO Pro and MobileNetV2-SSD, with calibration adaptation, alongside FOMO.

SAY THIS

We give you three architectures, so you pick the one that fits your accuracy/speed budget.

A real Linux/SBC runtime

The packaged .pxe runtime and UNOQ target are a first-class deployment path.

SAY THIS

A Raspberry Pi-class board or custom Linux SBC isn't a bolt-on for us — it's first-class.

Device catalog + compatibility check

69 boards, matched automatically, with on-device performance estimated before you build.

SAY THIS

You find out whether a model fits your board before you spend a build cycle finding out.

OTA fleet updates

Ship once, then push updates to devices already deployed in the field.

SAY THIS

Deployment isn't the end of the relationship — you can push improvements to devices already out there.

Data explorer built in

A labeled 2-D feature-space view (PCA) with click-to-inspect — not just a flat file list.

SAY THIS

You can see your dataset's structure before you waste a training run on it.

The demo walkthrough

Maps to the app's own sidebar, left to right — the order a prospect should click through.

1

DATA ACQUISITION & LABELING

Get their data in, fast

Upload images/CSV, or connect a device live. AI-assisted labeling and a labeling queue mean a prospect isn't hand-drawing every box.

"However your data shows up, it lands in the same place."

2

IMPULSE DESIGN

Show the pipeline, not a black box

Input → DSP block(s) → learning block, visibly chained.

“Every stage your data goes through is a block you can see and configure — nothing hidden inside one ‘train’ button.”

3

TRAINING

The architecture choice, not just one button

Pick from eight learning-block architectures, including the three detection trainers.

“You’re not stuck with whatever one architecture the vendor decided was good enough.”

4

EVALUATION & TESTING

Prove it before it ships

Variant comparison, live classification against a real device stream, and offline model testing with misclassification review.

“Watch it classify live on a real board, and see exactly which samples it got wrong, before production.”

5

DEPLOYMENT & DEVICES

Land it on their actual hardware

Pick a target, get a compatibility/performance check up front, build, and later push an OTA update.

“This is the step most platforms treat as an afterthought. For us it’s the point of the product.”

How to answer “how is this different from Edge Impulse?”

Honest positioning — don’t oversell breadth we don’t have yet; lead with what’s genuinely ours.

They’ll say	Say this back
“Edge Impulse has more input types and deploy targets”	True today for the widest long-tail of exotic targets — but ask what they actually need to ship. Our image pipeline is complete end-to-end, our device catalog covers 69 boards with real compatibility checking, and Motion (accelerometer/gyroscope) is in active development. We’re not chasing the longest checkbox list; we’re the platform you can actually run yourself.
“What about EON Tuner / auto-optimization?”	Not our current focus — we compete on three real detection architectures with calibration adaptation and a genuine on-device runtime story.
“Can our team collaborate / manage seats?”	Don’t answer this from the product — flag it to your deal desk. Multi-user/org support isn’t in the product today (see below).

■ **INTERNAL ONLY — DO NOT COMMIT THESE TO A PROSPECT**

These are real gaps against a mature platform like Edge Impulse Studio, tracked internally. If a prospect asks directly, say it's on the roadmap and loop in product — don't promise a date.

Multi-user / orgs / collaborators. Today it's single-owner per project — no team seats, roles, or shared-project permissions.

Plans, billing, usage limits. No metering or enforcement exists yet — nothing to quote here.

Input types beyond image. Image is the complete, proven pipeline. Motion (accelerometer/gyroscope) is in active development — real, but not finished. Audio is not yet reachable in the UI.

Deployment target breadth. The device catalog covers 69 boards for compatibility matching, but only a handful of deployment targets are UI-ready today, not the full embedded/MCU long tail.

Project versioning. “Versioning” today is a full project clone (a complete independent copy), not a lightweight snapshot/publish/restore flow — say “clone and compare,” not “version history.”

PetalEdge — an open-source edge ML platform for collecting data, training models, and deploying to edge devices. Built for internal enablement; verify current-quarter specifics with product before a customer commitment.